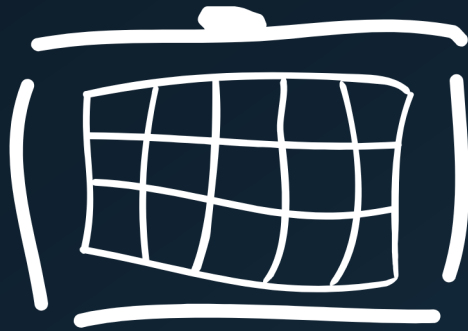




Radiator

CONFIGURATION GUIDE

Microsoft Entra ID App creation for Radiator Policy Server authentication

Date: 2026-08-26

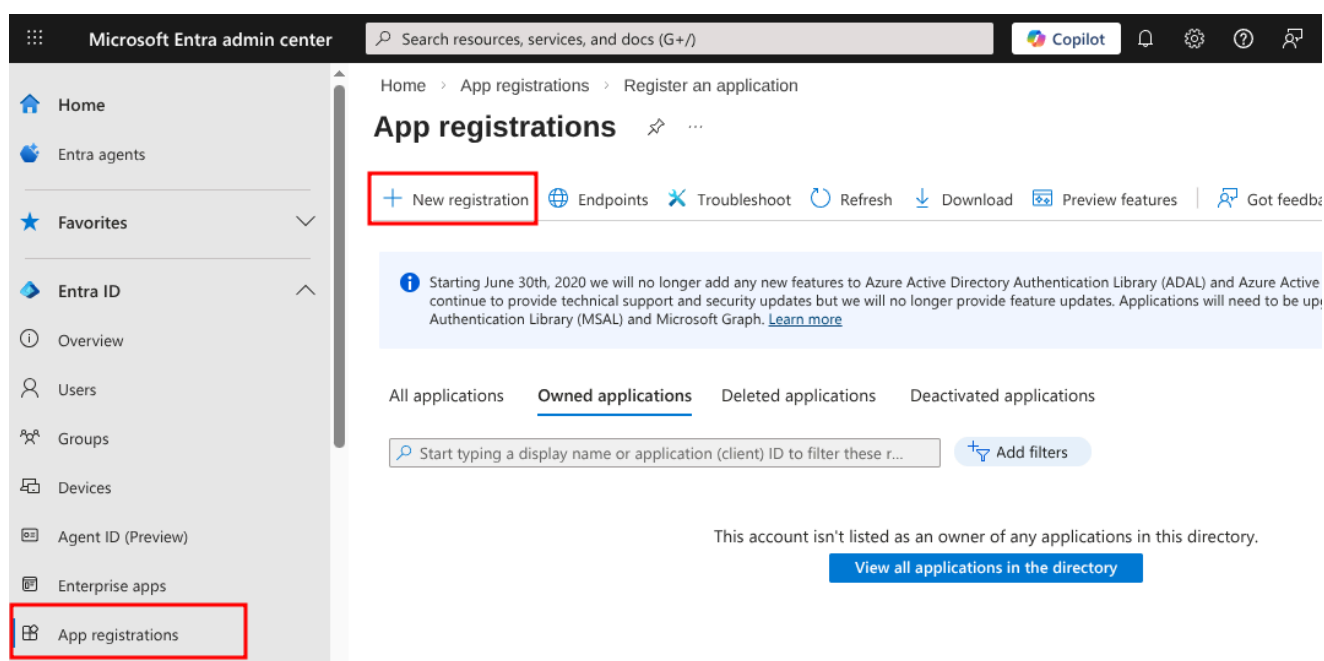
Creating an application to the Microsoft Entra ID is essential for establishing a trust relationship between Radiator Policy Server and the Microsoft identity platform. With the application you enable identity and access management (IAM) for Radiator Policy Server, allowing it to securely interact with Microsoft services and APIs.

The created trust is unidirectional. Radiator Policy Server trusts the Microsoft identity platform, and not the other way around. Once created, the application object can't be moved between different tenants.

Create the application

Create the app registration by following these steps:

1. Sign in to the [Microsoft Entra admin center](#) as a user with at least an *Application Developer* role
2. If you have access to multiple tenants, select the correct tenant.
3. Browse to **App registrations** and select **New registration**.
4. Enter a meaningful **Name**. App users can see this name, and it can be changed at any time. You can have multiple app registrations with the same name.
5. Select **Register** to complete the app registration.

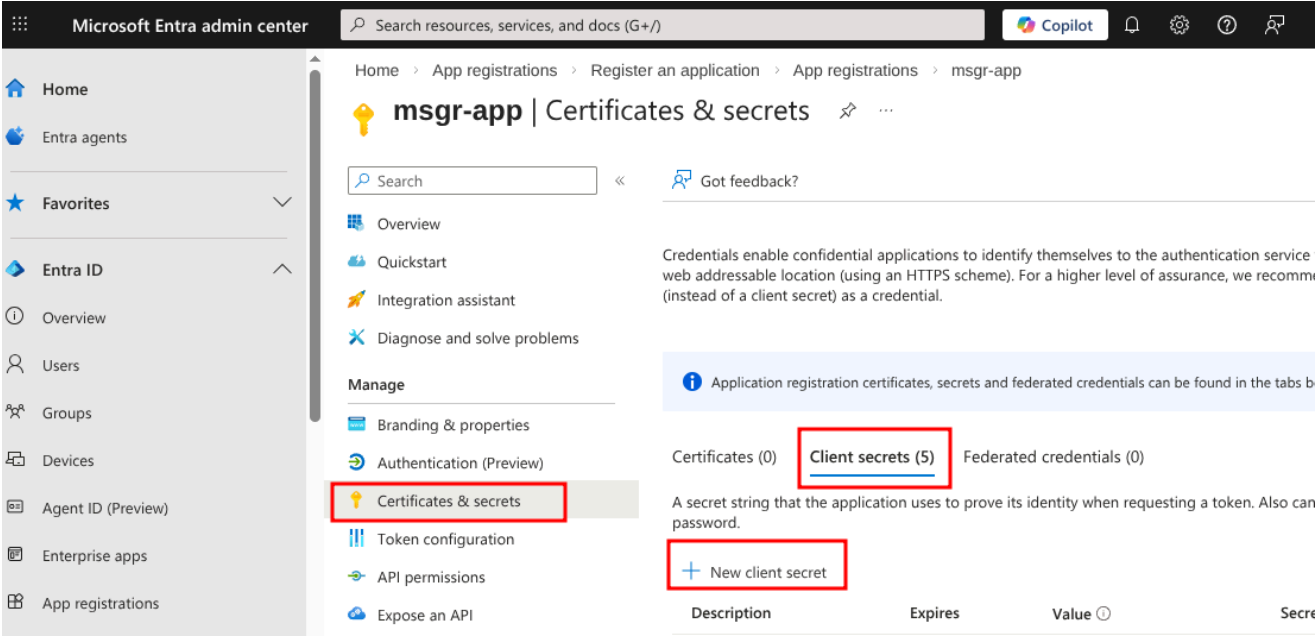


Add credentials

After registering an app you need to add client secrets (sometimes known as application password) as credentials. Credentials allow Radiator Policy Server to authenticate as itself, requiring no interaction from a user at runtime.

1. In the Microsoft Entra admin center, in **App registrations**, select your application.

2. Select **Certificates & secrets > Client secrets > New client secret**.



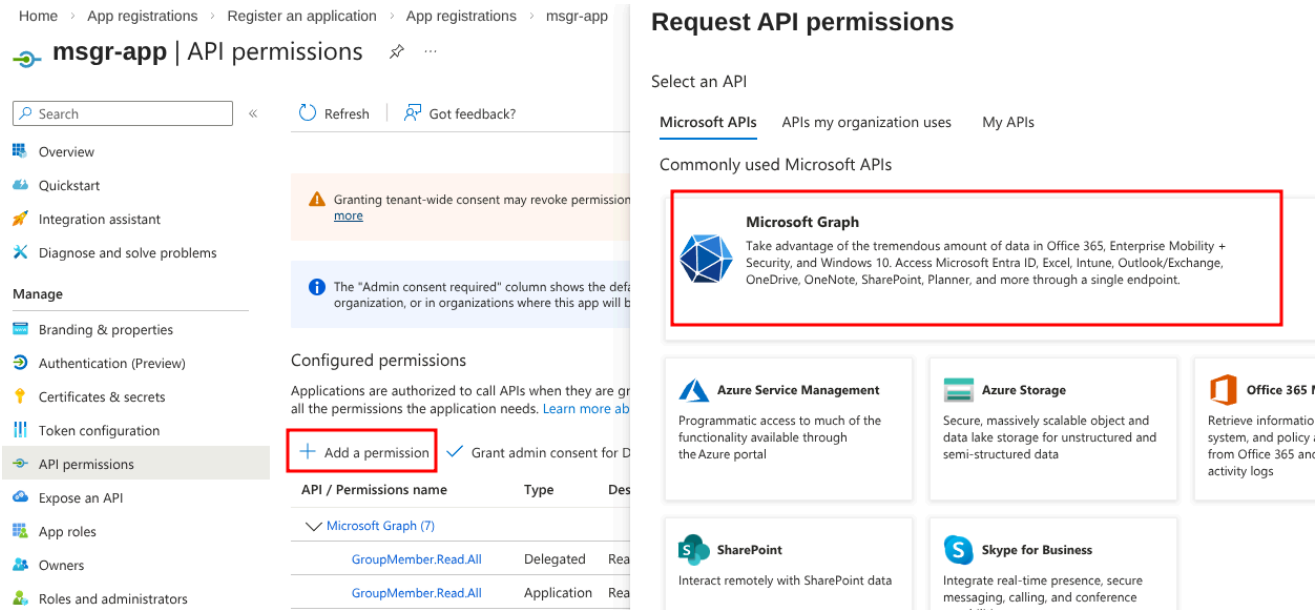
3. Add a description for your client secret.
4. Select an expiration for the secret or specify a custom lifetime.
5. Select **Add**.
6. *Record the secret's value* for use in Radiator Policy Server configuration (see [Radiator Entra ID configuration guide](#)). This secret value is *never displayed again* after you leave this page. If you lose the secret, you will have to create a new one.

Set permissions

Resource Owner Password Credentials flow

Radiator Policy Server also needs to access the data stored in Microsoft Graph. For Resource Owner Password Credentials (ROPC) flow configure delegated permission to Microsoft Graph to enable Radiator Policy Server to query the groups users belongs to as well as authenticating the user based on Microsoft Entra.

1. From the **Overview** page of your client application, select **API permissions > Add a permission > Microsoft Graph**



2. Select **Delegated permissions**

3. Under **Select permissions**, select the following permissions:

- GroupMember.Read.All
- openid
- profile
- User.Read

Home > App registrations > Register an application > App registrations > msgr-app

msgr-app | API permissions

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Granting tenant-wide consent may revoke permissions that have already been granted tenant-wide for that application. Permissions that users have already granted on their own behalf are affected. [Learn more](#)

The "Admin consent required" column shows the default value for an organization. However, user consent can be customized per permission, user, or app. This column may not reflect the value for an organization, or in organizations where this app will be used. [Learn more](#)

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. The list of configured permissions should include all the permissions the application needs. [Learn more about permissions and consent](#)

+ Add a permission ✓ Grant admin consent for Default Directory

API / Permissions name	Type	Description	Admin consent required	Status
Microsoft Graph (5)				
GroupMember.Read.All	Delegated	Read group memberships	Yes	Granted for Default Directory
offline_access	Delegated	Maintain access to data you have given it access to	No	Granted for Default Directory
openid	Delegated	Sign users in	No	Granted for Default Directory
profile	Delegated	View users' basic profile	No	Granted for Default Directory
User.Read	Delegated	Sign in and read user profile	No	Granted for Default Directory

4. Some permissions require an Admin grant. The **Grant admin consent for {your tenant}** button allows an admin to grant admin consent to the permissions configured for the application. When you select the button, a dialog is shown requesting that you confirm the consent action. After granting consent, the permissions that required admin consent are shown as having consent granted.

Client credentials flow

Permissions for Client credentials flow can be added to the same app used by the ROPC flow.

1. From the **Overview** page of your client application, select **API permissions** > **Add a permission** > **Microsoft Graph**

Home > App registrations > Register an application > App registrations > msgr-app

msgr-app | API permissions

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Granting tenant-wide consent may revoke permission [more](#)

The "Admin consent required" column shows the default value for an organization, or in organizations where this app will be used.

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+ Add a permission ✓ Grant admin consent for Default Directory

API / Permissions name	Type	Description	Admin consent required	Status
Microsoft Graph (7)				
GroupMember.Read.All	Delegated	Read group memberships	Yes	Granted for Default Directory
GroupMember.Read.All	Application	Read group memberships	Yes	Granted for Default Directory

Request API permissions

Select an API

Microsoft APIs APIs my organization uses My APIs

Commonly used Microsoft APIs

Microsoft Graph
Take advantage of the tremendous amount of data in Office 365, Enterprise Mobility + Security, and Windows 10. Access Microsoft Entra ID, Excel, Intune, Outlook/Exchange, OneDrive, OneNote, SharePoint, Planner, and more through a single endpoint.

Azure Service Management
Programmatic access to much of the functionality available through the Azure portal

Azure Storage
Secure, massively scalable object and data lake storage for unstructured and semi-structured data

Office 365
Retrieve information system, and policy information from Office 365 and activity logs

SharePoint
Interact remotely with SharePoint data

Skype for Business
Integrate real-time presence, secure messaging, calling, and conference

2. Select **Application permissions**

3. Under **Select permissions**, select the following permissions:

- GroupMember.Read.All
- User.Read.All

Home > App registrations > Register an application > App registrations > msggr-app

msggr-app | API permissions

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- Expose an API
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- Manifest

Support + Troubleshooting

- New support request

Granting tenant-wide consent may revoke permissions that have already been granted tenant-wide for that application. Permissions that users have already granted on their own behalf aren't [more](#)

The "Admin consent required" column shows the default value for an organization. However, user consent can be customized per permission, user, or app. This column may not reflect the value for an organization, or in organizations where this app will be used. [Learn more](#)

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. The list of configured permissions should include all the permissions the application needs. [Learn more about permissions and consent](#)

+ Add a permission Grant admin consent for Default Directory

API / Permissions name	Type	Description	Admin consent requ...	Status
Microsoft Graph (7)				
GroupMember.Read.All	Delegated	Read group memberships	Yes	Granted for Default Dire...
GroupMember.Read.All	Application	Read all group memberships	Yes	Granted for Default Dire...
offline_access	Delegated	Maintain access to data you have given it access to	No	Granted for Default Dire...
openid	Delegated	Sign users in	No	Granted for Default Dire...
profile	Delegated	View users' basic profile	No	Granted for Default Dire...
User.Read	Delegated	Sign in and read user profile	No	Granted for Default Dire...
User.Read.All	Application	Read all users' full profiles	Yes	Granted for Default Dire...

To view and manage consented permissions for individual apps, as well as your tenant's consent settings, try [Enterprise applications](#).

4. These permissions require an Admin grant. The **Grant admin consent for {your tenant}** button allows an admin to grant admin consent to the permissions configured for the application. When you select the button, a dialog is shown requesting that you confirm the consent action. After granting consent, the permissions that required admin consent are shown as having consent granted.

NOTE: It can take up to 60 minutes for the permissions to propagate into use.

Create exception for MFA requirement for the application

When authenticating with RADIUS protocol and ROPC or Clients credentials flow it is not possible to use the automatic multi-factor authentication, instead the MFA requirement must be disabled for the Microsoft Entra ID Application and trigger separately from Radiator Policy Server configuration. The recommended way to do this is with Conditional Access policies, as that will allow having MFA enabled for all other use cases. For example: MFA should only be disabled when the user is authenticating via the application created, not when the user is logging in to the Entra portal.

Using Conditional Access policies requires at minimum Microsoft Entra ID P1 licence and disabling organization security defaults. When disabling MFA via Conditional Access policy, users must not have Per-user MFA *enforced* or *enabled* in Entra portal (<https://entra.microsoft.com> > All users > Per-user MFA), instead the MFA should be enforced via Conditional Access policies for other use cases than with Radiator Policy Server authentication. In practice this means that MFA is not disabled/bypassed via Conditional Access policy, it is just not enforced to Radiator Policy Server authentication.

Example Conditional Access policy to require MFA except when using the application can be done for either all users or specific users, but the main thing is to exclude the created app when selecting Target resources. In the Grant section, "Require multifactor authentication" should be selected as control for granting the access:

Microsoft Entra admin center

Home > Conditional Access | Policies

Require MFA

Conditional Access policy

Delete View policy information View policy impact

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies. [Learn more](#)

Name *
Require MFA

Control access based on all or specific apps, agents, internet resources, actions, or authentication context. [Learn more](#)

Select what this policy applies to
Resources (formerly cloud apps)

Include **Exclude**

Select the resources to exempt from the policy

☐ None
☐ All internet resources with Global Secure Access
☐ All agent resources (Preview)
☒ Select resources

Select resources based on attributes
None

Select specific resources
msggr-app

Assignments

Users or agents (Preview)
All users included and specific users excluded

Target resources
All resources (formerly 'All cloud apps') included and 1 resource excluded

Network **NEW**
Not configured

Conditions
0 conditions selected

Access controls

Grant
1 control selected

Session
0 controls selected

To create a Conditional Access policy targeting members in your tenant with Global Secure Access (GSA) as a resource, make sure GSA is deployed in your tenant. [Learn more](#)

Recently, Microsoft has changed how the Conditional Access policies function when an application is excluded from a policy that otherwise targets All resources. Now the exclusion also takes into account which scopes Radiator uses with the application, and the behavior of the low-level baseline scopes has changed. All scopes are now evaluated and the Conditional Access policies are enforced regardless of the exclusion ([see more from Microsoft](#)).

To continue with the previous functionality, where baseline scopes were automatically excluded from policy enforcement, the Conditional Access Baseline scope enforcement must be changed to Customized behavior mode and another empty application must be registered so it can be excluded on the Conditional Access policy rule as shown in [Microsoft instructions](#).

Microsoft Entra admin center

Search resources, services, and docs (G+)



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Conditional Access

Multifactor authentication

Identity Secure Score

Home > App registrations

Register an application

* Name

The user-facing display name for this application (this can be changed later).

Baseline scopes customization

Supported account types

Choose the account types that can use this application or access this API

Single tenant only - Default Directory

Help me choose

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Select a platform

e.g. https://example.com/auth

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by addi

By proceeding, you agree to the Microsoft Platform Policies

Register

Microsoft Entra admin center

Search resources, services, and docs (G+)

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Conditional Access

Home > App registrations > Conditional Access | Policies

Require MFA

Conditional Access policy

Delete

View policy information

View policy impact

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies.
[Learn more](#)

Name *

Require MFA

Assignments

Users or agents ¹

All users included and specific users excluded

Target resources ¹

All resources (formerly 'All cloud apps') included and 2 resources excluded

Network ¹

Not configured

Conditions ¹

0 conditions selected

Access controls

Grant * ¹

1 control selected

Control access based on all or specific apps, agents, internet resources, actions, or authentication context. [Learn more](#)

Select what this policy applies to

Resources (formerly cloud apps)

Include

Exclude

Select the resources to exempt from the policy

None

All internet resources with Global Secure Access

All agent resources ¹

Select resources

Select resources based on attributes ¹

None

Select specific resources ¹

msgr-app and 1 more

BS

Baseline scopes customization

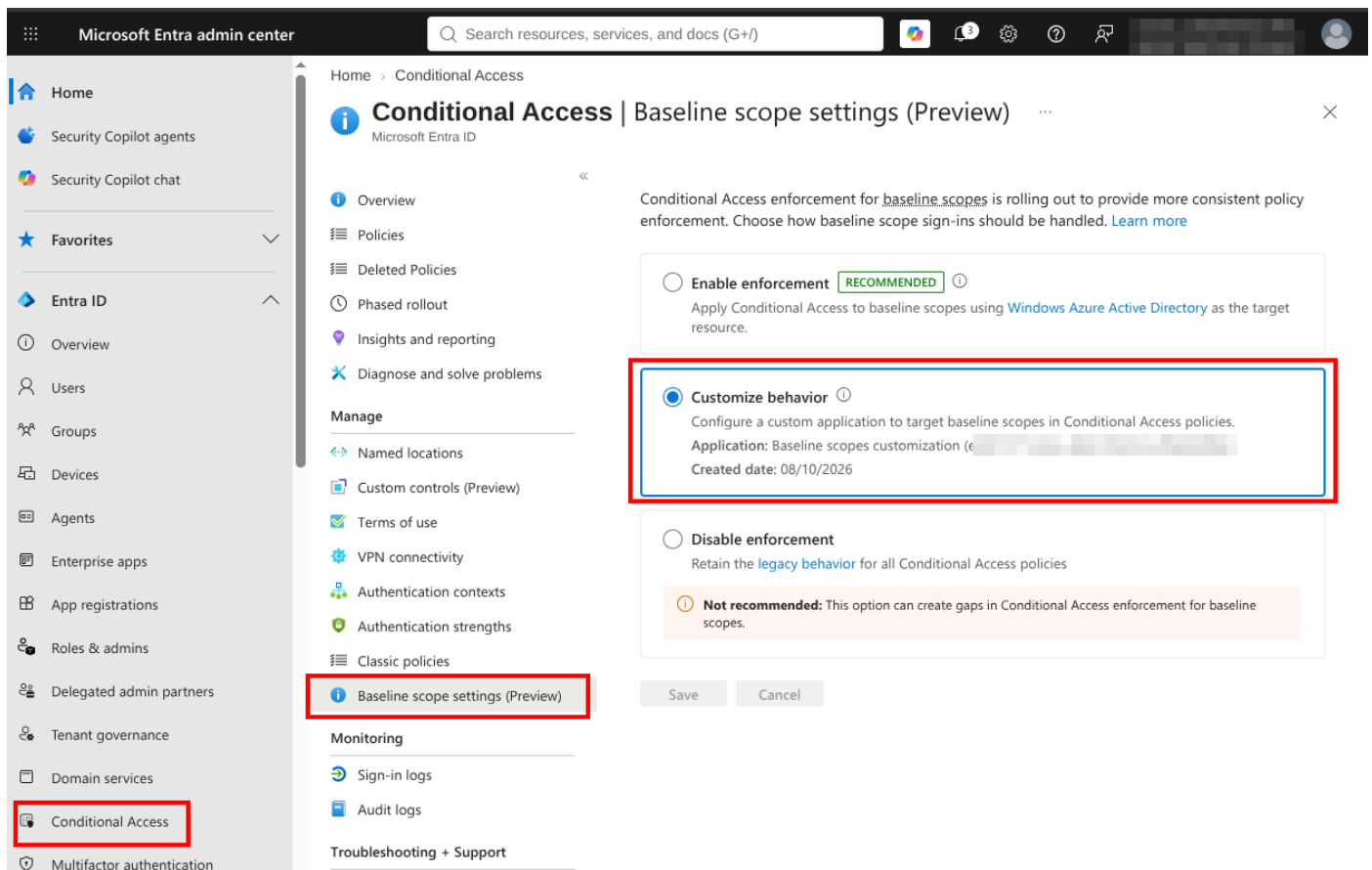
M

msgr-app

Configuration guide


Radiator

Document date: 2026-08-26



NOTE: Do not use the Radiator application on the Baseline scope customized behavior configuration as it has various permissions configured.

It is also possible to exclude Radiator from MFA requirement by excluding the users or user groups, or by selecting the other applications to the policy individually without using All resources. Excluding the users/user groups means that MFA is not required for the user at all, regardless of where they log in. Selecting all the applications except the Radiator application means the list may need to be manually updated from time to time to catch possible new Microsoft or other applications.

Radiator Policy Server configuration

After completing the previous steps you will need to update your Radiator Policy Server configuration. See Entra ID configuration guide ([Radiator Entra ID configuration guide](#)) for more details.

Troubleshooting

Here is a collection of known errors shown on the Radiator Policy Server authentication log, and how they can be fixed either on Radiator Policy Server configuration or on Microsoft Entra ID. When Radiator Policy Server configuration is edited, remember to restart Radiator Policy Server after changes

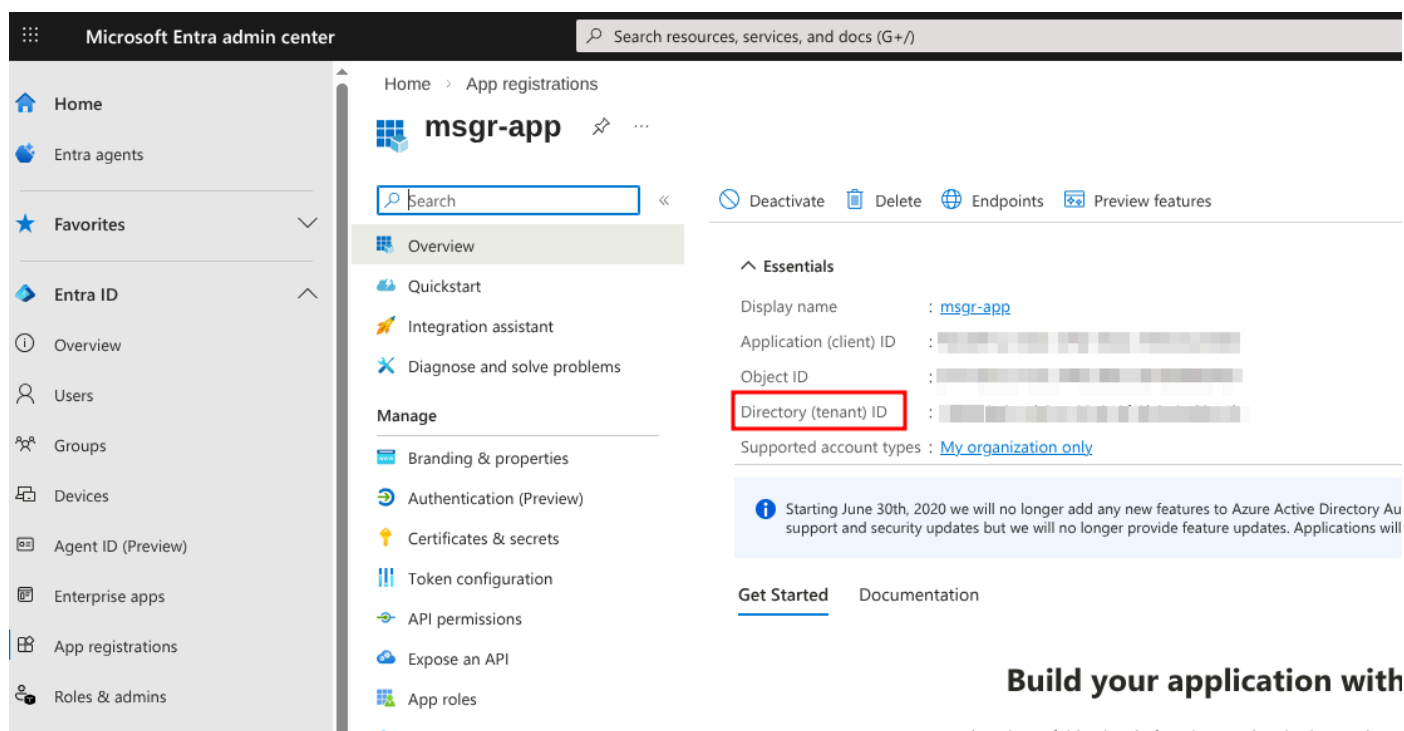
Error: Acquiring access token failed

```
{"error": "invalid_request", "error_description": "AADSTS90002: Tenant
```

'xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx' not found. Check to make sure you have the correct tenant ID and are signing into the correct cloud. Check with your subscription administrator, this may happen if there are no active subscriptions for the tenant.

Solution

Check that Radiator Policy Server configuration parameter `vars.aaa.oauth2.token-url` is correct, it should contain the tenant ID and not for example application ID



Error Acquiring access token failed

`{"error":"invalid_grant","error_description":"AADSTS65001: The user or administrator has not consented to use the application with ID 'xxxxxxxx-xxxx-xxxx-xxxxxxxxxxxx' named 'msgr-app'. Send an interactive authorization request for this user and resource."}`

Solution

There are multiple solutions to this error:

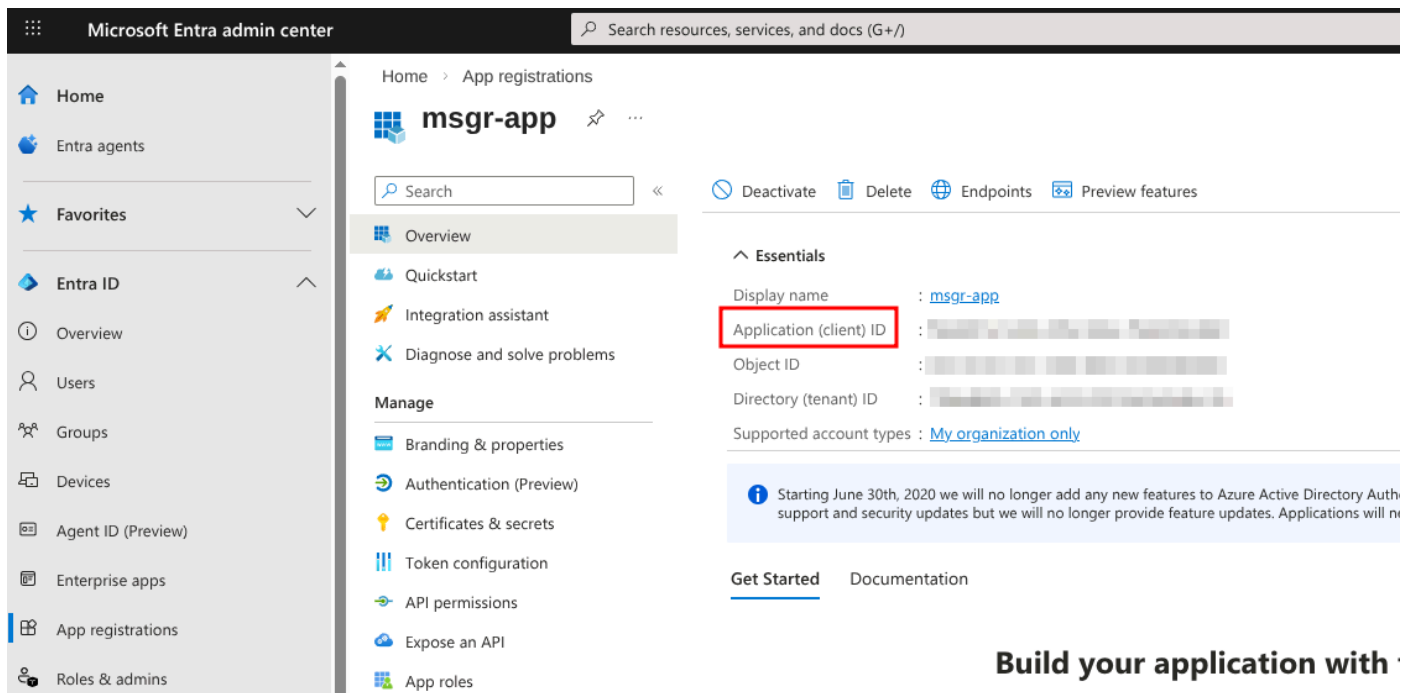
- Check the Microsoft Entra ID application permissions are as listed in "Set permissions" chapter
- Check administrator consent has been granted as shown in "Set permissions" chapter

Error: Acquiring access token failed

`{"error":"unauthorized_client","error_description":"AADSTS700016: Application with identifier 'xxxxxxxx-xxxx-xxxx-xxxxxxxxxxxx' was not found in the directory 'Default Directory'. This can happen if the application has not been installed by the administrator of the tenant or consented to by any user in the tenant. You may have sent your authentication request to the wrong tenant. Trace ID: 2bb28fbb-82ea-4365-b956-d88ad0d21d00"}`

Solution

Check that Radiator Policy Server configuration and the `vars.aaa.oauth2.client` parameter has the correct application ID



Error: Acquiring access token failed

```
{ "error": "invalid_grant", "error_description": "AADSTS50126: Error validating credentials due to invalid username or password" }
```

Solution

Check the given password is correct

Error: Acquiring access token failed

```
{ "error": "invalid_grant", "error_description": "AADSTS50034: The user account {EUII Hidden} does not exist in the xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx directory. To sign into this application, the account must be added to the directory" }
```

Solution

Check the username used is correct and exists in the Microsoft Entra ID

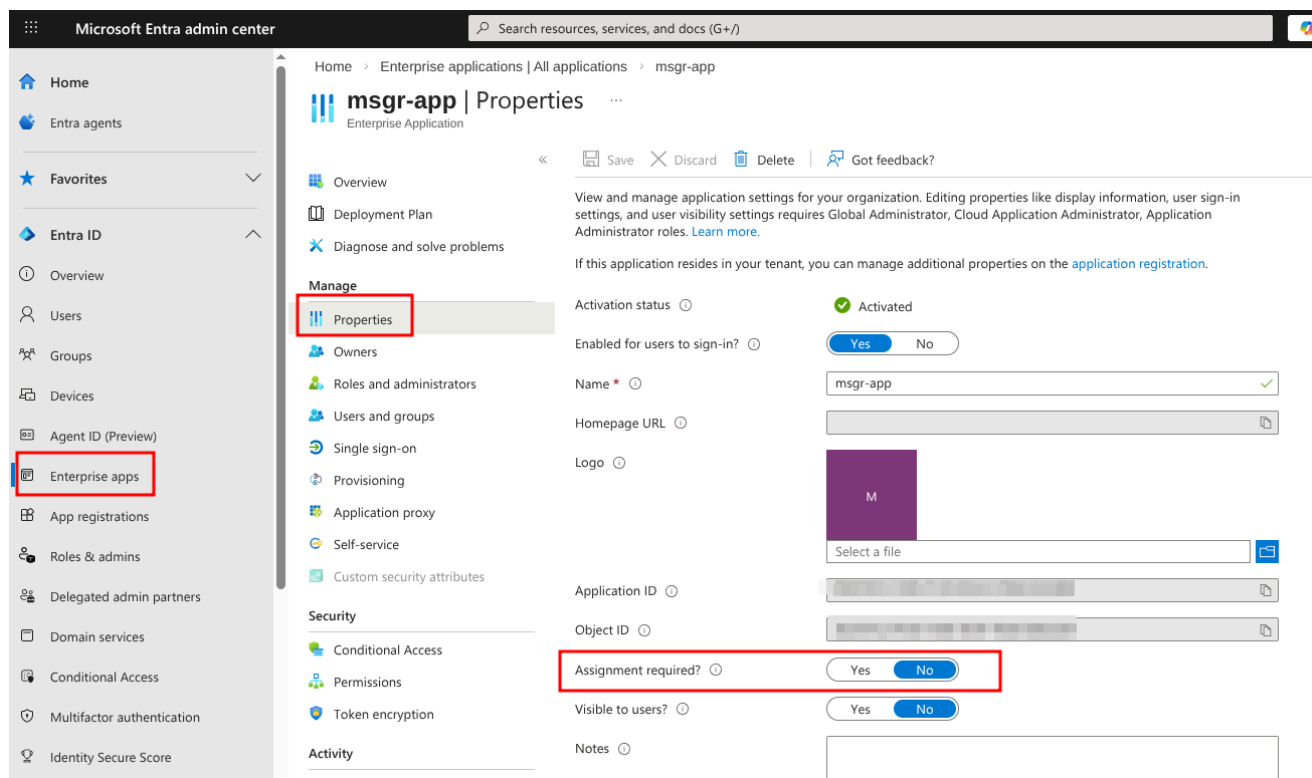
Error: Acquiring access token failed

```
{ "error": "invalid_grant", "error_description": "AADSTS50105: Your administrator has configured the application msgr-ent ('xxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx') to block users unless they are specifically granted ('assigned') access to the application. The signed in user '{EUII Hidden}' is blocked because they are not a direct member of a group with access, nor had access directly assigned by an administrator. Please contact your administrator to assign access to this application." }
```

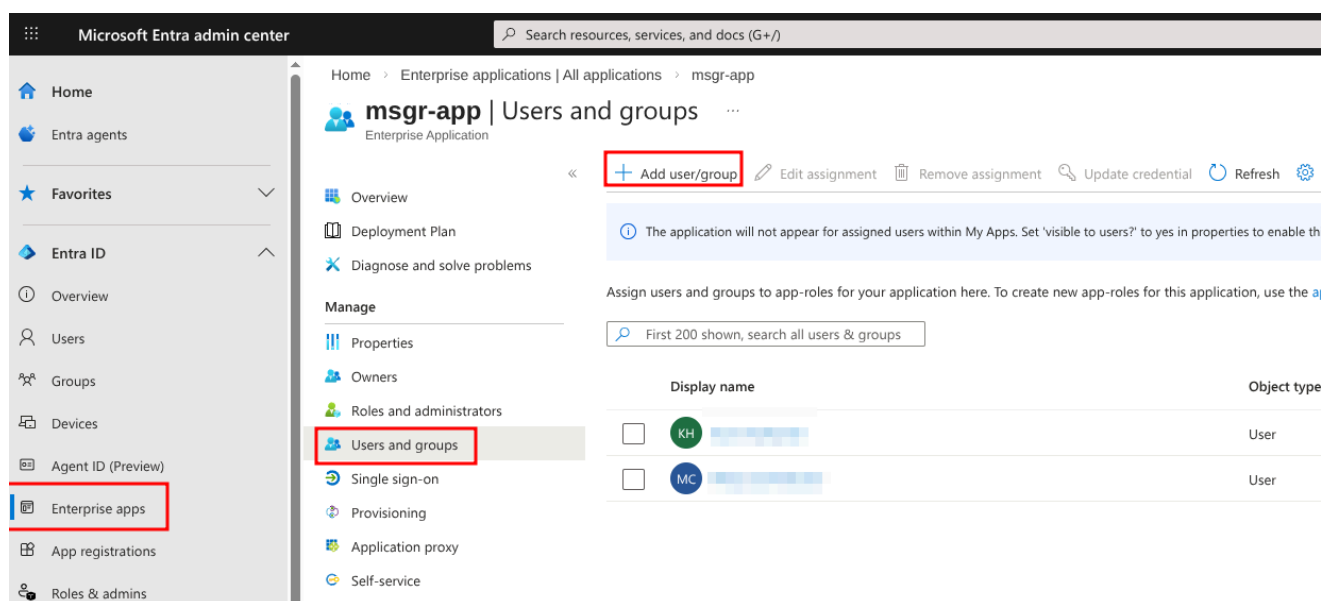
Solution

This error is usually only appearing if the application has been created as an enterprise application and the Assignment required has been set to yes. There are multiple solutions to this error:

- Check the Microsoft Entra ID application permissions are as listed in "Set permissions" chapter
- Check administrator consent has been granted as shown in "Set permissions" chapter
- Set the Assignment required option as No



- If the Assignment required cannot be set to No, then check the user has been assigned access either directly or via group

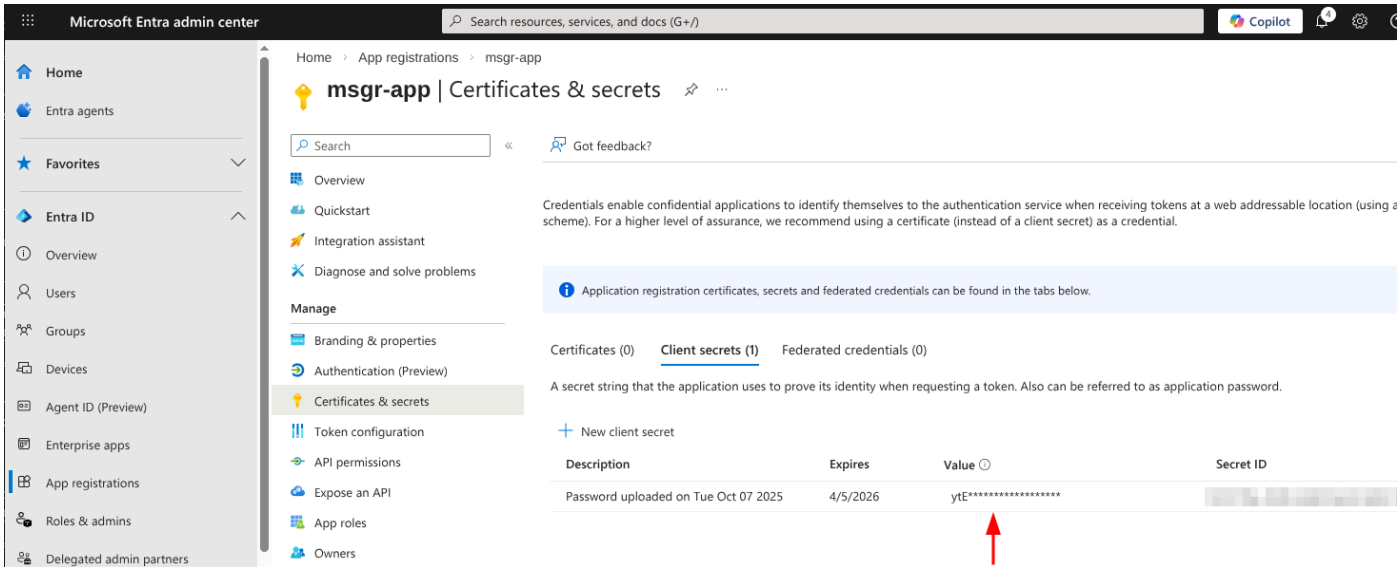


Error: Acquiring access token failed

`{"error": "invalid_client", "error_description": "AADSTS7000215: Invalid client secret provided. Ensure the secret being sent in the request is the client secret value, not the client secret ID, for a secret added to app 'xxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxx'"}`

Solution

Check the Radiator Policy Server configuration and the `vars.aaa.oauth2.secret` parameter has the secret value (not secret ID) from the correct application.

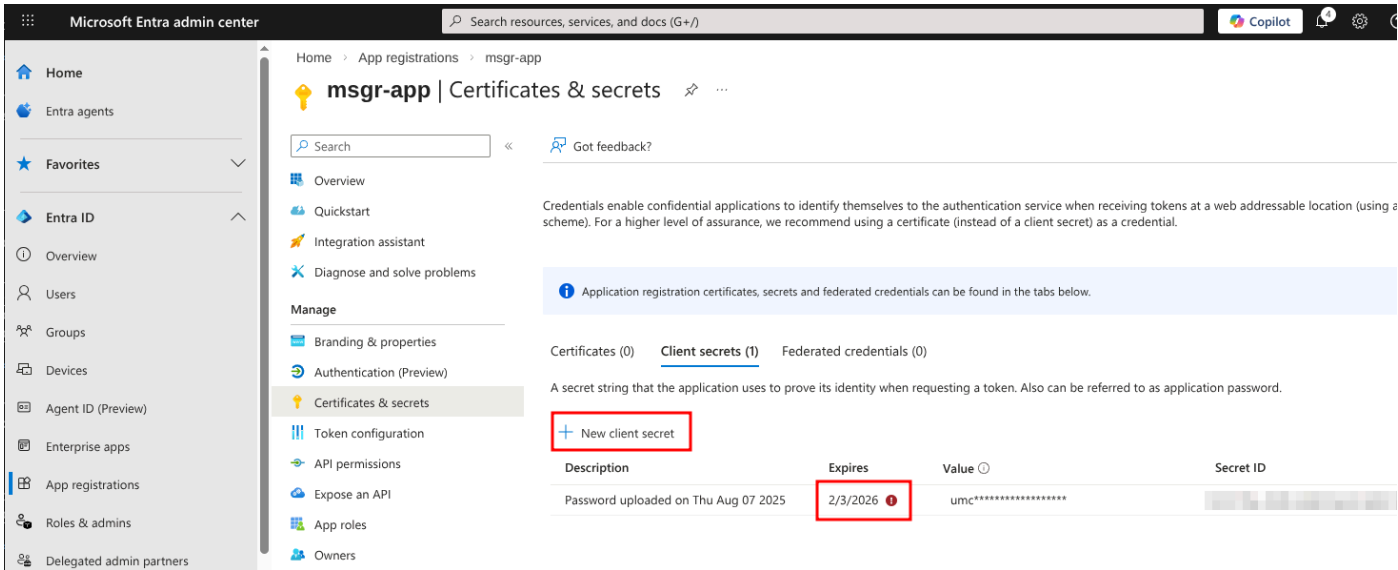


Error: Acquiring access token failed

`{"error": "invalid_client", "error_description": "AADSTS7000222: The provided client secret keys for app 'xxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx' are expired. Visit the Azure portal to create new keys for your app: https://aka.ms/NewClientSecret, or consider using certificate credentials for added security: https://aka.ms/certCreds."}`

Solution

The secret defined on the Microsoft Entra ID has expired. Create a new secret and update it to the Radiator Policy Server configuration `vars.aaa.oauth2.secret` parameter.



Error: Acquiring access token failed

`{"error": "invalid_grant", "error_description": "AADSTS50079: Due to a configuration change made by your administrator, or because you moved to a new location, you must enroll in multi-factor authentication to access '00000003-0000-0000-c000-000000000000'."}`

Solution

Microsoft Entra ID administrator has enforced the multi-factor authentication for the user trying to authenticate, the Security defaults are enabled or Conditional Access Baseline scope settings are not customized. Possible solutions are:

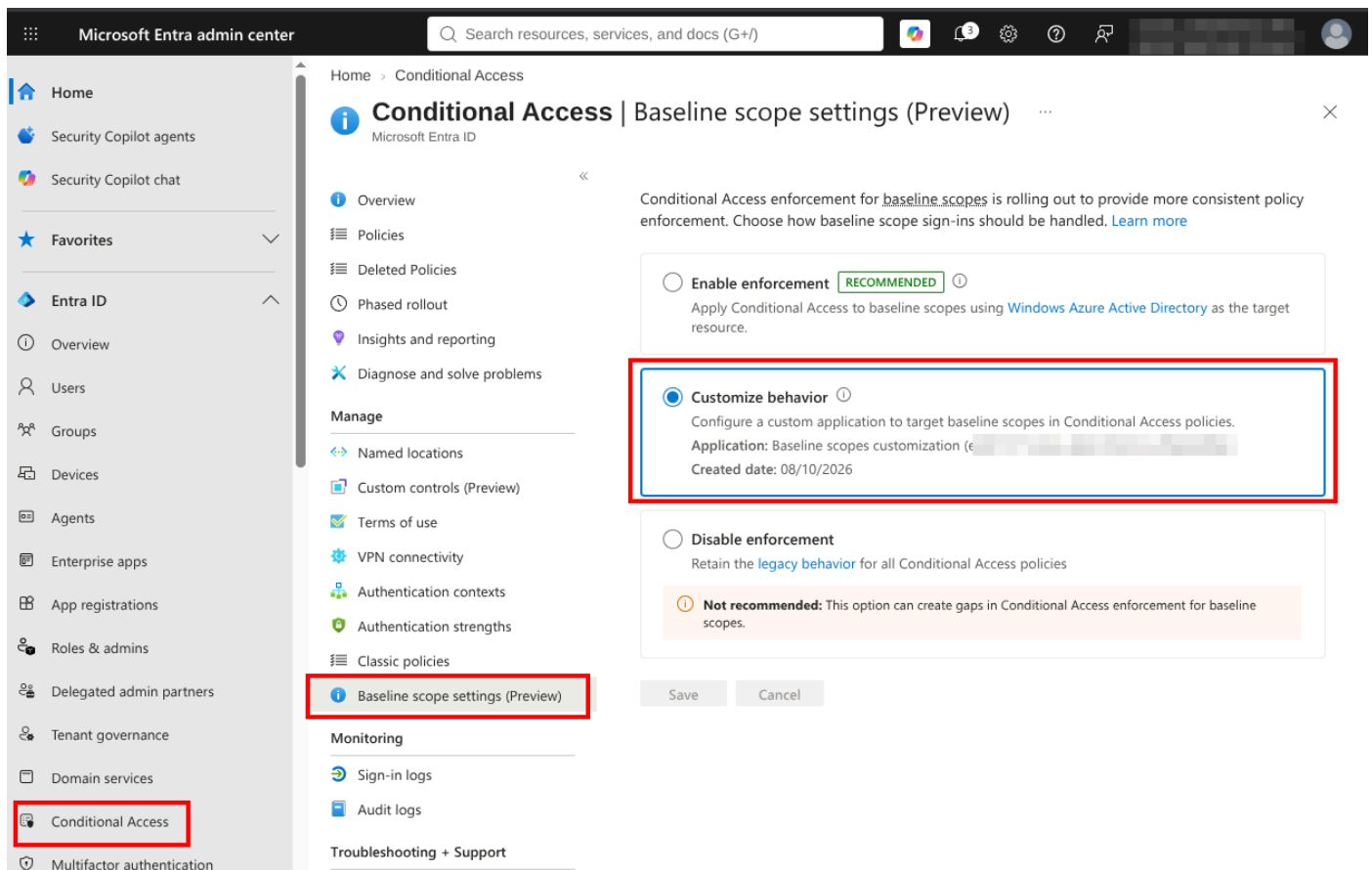
- Disable the multi-factor authentication for the user from the Entra portal <https://entra.microsoft.com> > All users > Per-user MFA
- Disable the multi-factor authentication from all users <https://entra.microsoft.com> > Overview > Properties > Security defaults and take Conditional Access Policies into use
- Disable multi-factor authentication requirement from Conditional Access policy <https://entra.microsoft.com> > Conditional Access or create app passwords for the user: <https://learn.microsoft.com/en-us/entra/identity/authentication/howto-mfa-app-passwords>
- Use Customized behavior on Conditional Access Baseline scope settings: <https://learn.microsoft.com/en-us/entra/identity/conditional-access/concept-enforcement-resource-exclusions#customize-behavior>

Using Conditional Access policies requires at minimum Microsoft Entra ID P1 licence and disabling organization security defaults. When disabling MFA via Conditional Access policy, users must not have Per-user MFA *enforced* or *enabled* in Entra portal (<https://entra.microsoft.com> > All users > Per-user MFA), instead the MFA should be enforced via Conditional Access policies for other use cases than with Radiator Policy Server authentication. In practice this means that MFA is not disabled/bypassed via Conditional Access policy, it is just not enforced to Radiator Policy Server authentication.

Example Conditional Access policy to require MFA except when using the application can be done for either all users or specific users, but the main thing is to exclude the created app when selecting Target resources. In the Grant section, "Require multifactor authentication" should be selected as control for granting the access. In the same policy, exclude also the application used for the Baseline scope customization.

The screenshot displays the Microsoft Entra admin center interface. The left-hand navigation pane shows the 'Conditional Access' option highlighted with a red box. The main content area shows the configuration for a policy named 'Require MFA', which is also highlighted with a red box. The policy is configured with the following settings:

- Name:** Require MFA
- Assignments:** All users included and specific users excluded
- Target resources:** All resources (formerly 'All cloud apps') included and 2 resources excluded (highlighted with a red box)
- Network:** Not configured
- Conditions:** 0 conditions selected
- Access controls:** 1 control selected (Require multifactor authentication)
- Grant:** 1 control selected
- Select what this policy applies to:** Resources (formerly cloud apps)
- Include/Exclude:** Exclude (highlighted with a red box)
- Select the resources to exempt from the policy:** Select resources (radio button selected)
- Select resources based on attributes:** None
- Select specific resources:** msgr-app and 1 more
- Baseline scopes customization:** BS (highlighted with a red box)
- msgr-app:** M (highlighted with a red box)



Error: Acquiring access token failed

`{"error": "invalid_grant", "error_description": "AADSTS53003: Access has been blocked by Conditional Access policies. The access policy does not allow token issuance."}`

Solution

Check the Conditional Access Policies on Microsoft Entra ID and allow access for example to the target application

Error: Acquiring access token failed

`{"error": "unauthorized_client", "error_description": "AADSTS7000112: Application 'xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx'(msgr-app) is disabled"}`

Solution

Check that the Enterprise Application Properties allow users to sign in with the app from the Microsoft Entra ID

Acquiring access token failed: error sending request for url

Acquiring access token failed: error sending request for url (<https://login.microsoftonline.com/<tenant-ID>/oauth2/v2.0/token>): operation timed out

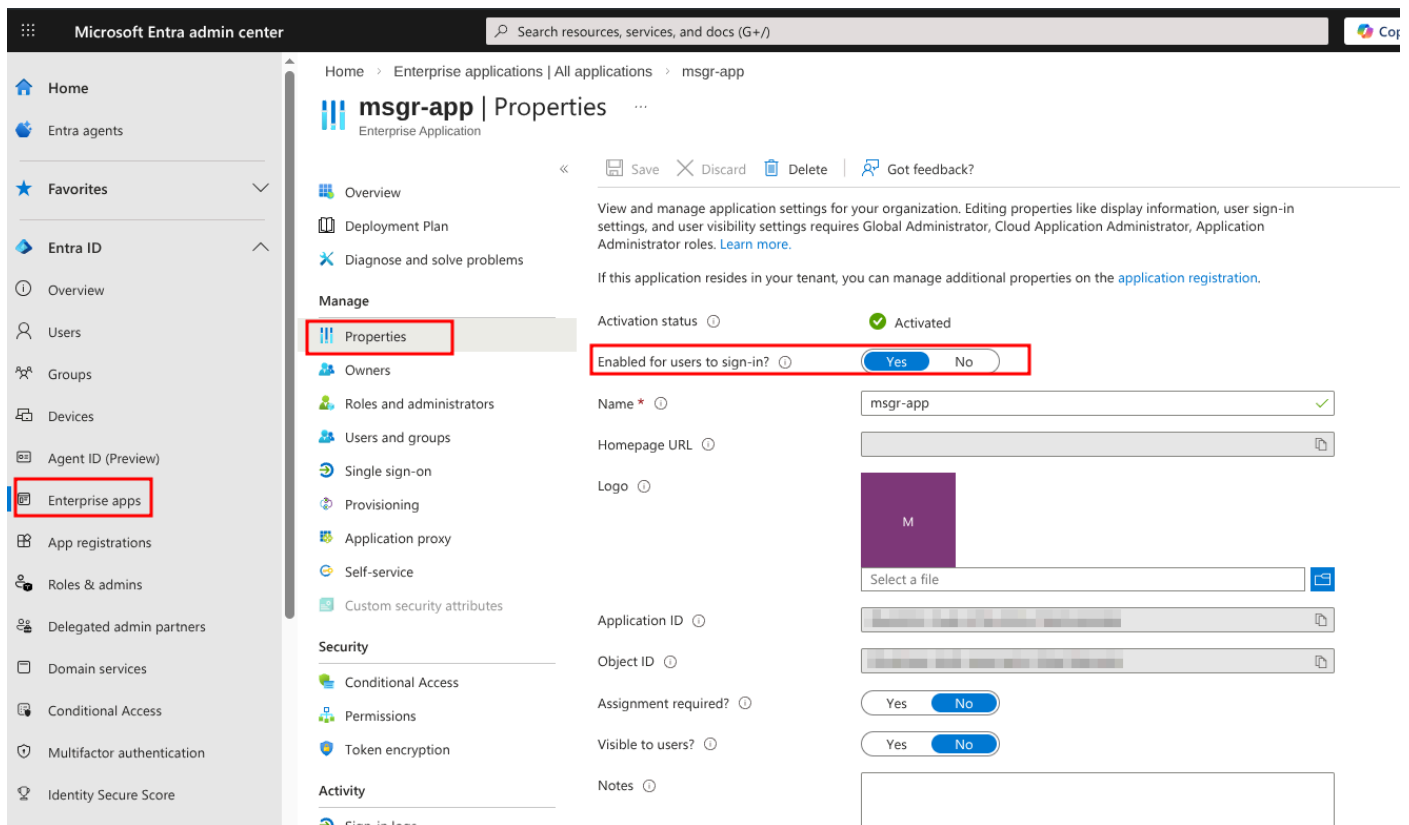
Additionally there are no sign-in tries visible on the Microsoft Entra Sign-in logs.

Solution

This error can happen for example if Microsoft Entra ID P2 license has been enabled and related to it ApplicationManagement does automatic service principal updates. These updates are visible for example on the Microsoft Entra ID directory Audit logs:

2/23/26, 2:58:00 PM	Core Directory	ApplicationManagement	Update service principal	Success
2/23/26, 2:57:57 PM	Core Directory	ApplicationManagement	Update service principal	Success
2/23/26, 2:57:56 PM	Core Directory	ApplicationManagement	Update service principal	Success

To get the application functionality back, try to disable and enable the Enterprise Application from Properties:



Error: No matching realm found

Default Handler: No matching realm found.

Solution

Check that Radiator Policy Server configuration and the username trying to authenticate have the same realm: `user@realm` and handler with condition like `aaa.identity.realm == "realm"`;

Error: None of user groups matched

None of user groups matched, so the authentication is rejected

Solution

Radiator Policy Server configuration file `50_policies.conf` defines groups that the user should belong to. This group membership is checked from the Microsoft Entra ID, and if the user does not belong to any of the groups listed on the Radiator Policy Server configuration, the above error is shown.